

RIO DE JANEIRO ALFONSOS, Brazil

WMO: 837480

Lat: 22.875S

Lon: 43.385W

Elev: 110

StdP: 14.64

Time Zone: -3.00 (SAB)

Period: 98-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	55.0	57.0	49.8	53.3	67.9	51.8	57.3	67.8	17.3	76.9	14.8	76.0	1.9	240	0.282

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	20.0	98.9	76.9	96.9	76.5	95.0	76.5	81.5	91.3	80.3	90.2	79.2	88.5	7.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.0	151.2	87.1	77.4	143.2	85.0	76.9	140.8	84.3	45.3	91.4	44.0	90.5	42.8	88.2	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.4	12.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.1	82.2	83.5	80.8	76.3	72.0	70.4	70.2	71.7	74.0	76.0	77.1	80.1		
	DBStd	6.31	4.69	3.98	4.62	4.13	3.98	4.07	4.14	4.27	4.94	4.58	5.28	4.90		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	21	0	0	0	0	2	6	7	4	2	0	0	0		
	CDD50	9537	997	937	954	788	681	613	626	671	720	805	813	932		
	CDD65	4084	532	517	489	338	218	169	168	211	272	340	363	467		
	CDH74	37404	5892	6195	4891	2567	1466	1068	1217	1675	2108	2661	3031	4633		
	CDH80	16117	2906	3210	2128	873	444	314	395	626	879	1059	1231	2052		
(14) Wind	WSAvg	4.7	5.0	5.0	4.6	4.3	3.9	3.6	3.8	4.6	5.2	5.6	5.6	5.2		
(15) Precipitation	PrecAvg	52.8	8.5	5.4	6.5	3.9	2.7	1.9	1.6	1.5	3.1	3.8	6.2	8.0		
	PrecMax	73.7	14.9	12.3	14.7	10.4	5.2	7.9	4.5	5.3	5.9	9.8	13.1	18.8		
	PrecMin	22.8	2.4	0.3	1.4	0.8	1.1	0.1	0.2	0.4	0.9	0.9	1.7	2.9		
	PrecStd	11.7	3.3	3.0	3.6	2.2	1.2	1.4	1.1	1.1	1.5	2.3	2.8	3.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.6	102.1	99.0	95.1	92.9	91.1	91.5	93.6	98.4	98.2	98.9	100.0		
		MCWB	76.7	77.4	78.9	74.8	73.9	69.9	70.3	69.8	70.8	73.6	75.4	76.6		
	2%	DB	98.2	98.8	96.5	91.0	88.2	87.4	87.8	90.0	93.1	93.2	94.9	96.3		
		MCWB	77.6	77.0	78.2	75.0	71.8	70.2	68.9	69.6	71.5	73.2	75.4	77.2		
	5%	DB	95.3	96.6	93.3	87.9	84.6	84.0	84.5	86.7	89.3	89.5	91.2	93.3		
		MCWB	77.3	77.0	77.7	74.9	70.9	69.4	68.7	69.5	71.1	73.1	75.1	77.0		
	10%	DB	93.0	93.6	89.9	84.8	81.0	80.4	80.9	82.7	84.6	86.0	87.4	89.9		
		MCWB	77.0	76.8	77.1	74.0	69.4	69.0	68.9	69.1	70.6	72.7	74.6	76.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.6	83.3	83.1	80.3	77.1	74.7	75.5	76.1	76.9	77.5	79.0	81.6		
		MCDB	93.5	94.1	93.5	87.6	88.1	83.7	84.6	86.7	88.3	88.4	90.9	91.8		
	2%	WB	80.2	81.8	81.1	77.9	74.5	72.5	72.5	73.7	74.5	76.0	77.5	79.9		
		MCDB	92.0	91.1	90.3	84.9	82.6	81.3	80.6	83.2	85.5	86.6	88.6	89.8		
	5%	WB	79.0	80.4	79.7	76.3	72.3	71.0	71.0	71.7	73.1	74.7	76.4	78.7		
		MCDB	89.9	89.8	88.1	83.8	81.0	79.0	78.8	80.6	83.0	85.1	86.5	88.1		
	10%	WB	78.1	79.0	78.6	75.0	70.8	69.9	69.7	70.0	71.9	73.5	75.3	77.5		
		MCDB	88.2	87.9	86.4	83.0	79.4	77.1	77.4	78.9	80.9	82.9	84.2	86.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.5	20.0	16.8	17.2	18.7	19.0	20.0	21.0	19.2	17.5	15.9	16.8		
		MCDBR	23.3	23.7	21.3	20.9	24.9	25.8	27.6	28.9	28.4	24.7	22.9	22.3		
	5% WB	MCWBR	7.5	6.9	7.1	8.3	10.1	11.1	11.9	11.2	9.2	8.8	7.4	7.6		
		MCWBR	19.8	18.5	17.3	17.5	21.2	19.7	21.4	22.5	21.0	19.9	19.0	18.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.413	0.410	0.409	0.399	0.382	0.381	0.382	0.414	0.494	0.469	0.434	0.425		
		taud	2.470	2.461	2.472	2.487	2.508	2.498	2.476	2.367	2.119	2.252	2.378	2.423		
	Ebn at Noon	Ebn at Noon	296	292	284	273	265	259	262	265	255	272	288	293		
		Edh at Noon	38	37	36	32	29	28	30	36	50	46	41	40		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1906	1982	1693	1486	1180	1102	1141	1354	1455	1635	1682	1861		
	RadStd	RadStd	280	204	141	107	64	82	97	104	187	193	139	168		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	-3.04	N/A	-2.52	-2.99	N/A	N/A	N/A	N/A

Nomenclature: See separate page